

Homework 3

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1 Answers

1.

$$\begin{aligned}y_i &= e^\alpha \delta^{d_i} z_i^\gamma e^{\eta_i} \\ \Rightarrow \ln(y_i) &= \ln(e^\alpha \delta^{d_i} z_i^\gamma e^{\eta_i}) \\ \Rightarrow \ln(y_i) &= \alpha + \ln(\delta) d_i + \gamma \ln(z_i) + \eta_i\end{aligned}$$

2. δ captures the proportional change in electricity usage followed by a home being assigned to the retrofitting program ($d_i = 1$).

3. When a home has ben retrofitted ($d_i = 1$):

$$y_i(1) = e^\alpha \delta z_i^\gamma e^{\eta_i}$$

When a home has not received the retrofit ($d_i = 0$):

$$y_i(0) = e^\alpha z_i^\gamma e^{\eta_i}$$

$$\begin{aligned}\Rightarrow \Delta y_i &= y_i(1) - y_i(0) \\ \Rightarrow \Delta y_i &= e^\alpha z_i^\gamma e^{\eta_i} (\delta - 1) \\ \Rightarrow \Delta y_i &= e^\alpha \delta^{d_i} z_i^\gamma e^{\eta_i} (\delta - 1) / \delta^{d_i} \\ \Rightarrow \Delta y_i &= y_i (\delta - 1) / \delta^{d_i}\end{aligned}$$

Since $\Delta d_i = 1 - 0 = 1$:

$$\Delta y_i / \Delta d_i = y_i (\delta - 1) / \delta^{d_i}$$

$\Delta y_i / \Delta d_i$ tells us the change in electricity usage of a home when it receives the retrofit.

4.

$$\begin{aligned}\delta \ln(y_i) / \delta z_i &= \frac{1}{y_i} \frac{\delta y_i}{\delta z_i} = \frac{\gamma}{z_i} \\ \Rightarrow \frac{\delta y_i}{\delta z_i} &= \gamma \frac{y_i}{z_i}\end{aligned}$$

When z_i is the size of the home in square feet, $\frac{\delta y_i}{\delta z_i}$ shows the change in electricity usage due to a change in the size of a home.

5. Table 1 shows the regression results along with the coefficient estimates of the average marginal effects.

6. Figure 1 shows the average marginal effects of outdoor temperature and square feet of the home with their bootstrapped confidence intervals

2 Tables

Table 1: OLS Regression Results and Average Marginal Effects Bootstrapped with 1000 Replications.

Variable [ln(x)]	OLS Estimate	Average Marginal Effect
Constant	-0.768728 [-1.876, 0.349]	
Retrofit	-0.100547 [-0.113, -0.089]	0.628662 [0.618, 0.641]
Square feet of home	0.894291 [0.880, 0.910]	3.997015 [0.446, 7.485]
Temperature	0.281345 [0.032, 0.528]	-113.975304 [-127.942, -100.624]

Note: The dependent variable for the OLS regression is $\ln(\text{electricity})$. Average marginal effects are the mean of $\frac{dy}{dx}$ where y is household electricity consumption in that month. Confidence intervals in parantheses.

3 Figures

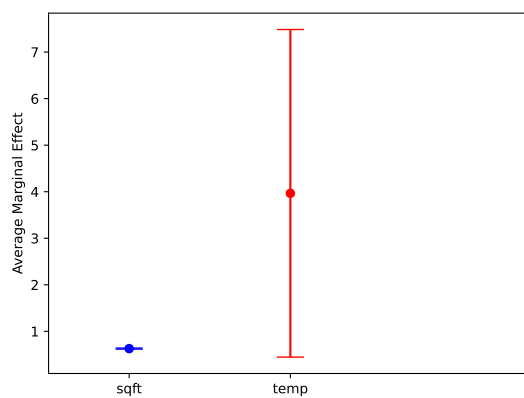


Figure 1: Average marginal effect estimates with 95% confidence intervals bootstrapped using 1000 replications.